

INFLECTION AND THE LEXEME*

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Abstract

The notion of 'lexeme' is central to realizational theories of morphology and to the notion of 'morphology by itself'. It is generally assumed that 'inherent' inflections such as Plural or Past Tense impart a meaning to the inflected word. However, this runs counter to the usual understanding of the notion 'lexeme', which is supposed to have a single constant meaning for all word forms. Since derivational morphology is supposed to create new lexemes by adding a new component of meaning this makes it difficult to distinguish inflection from derivation, which in turn undermines the whole lexeme concept. The problem evaporates if we assume that syntactic features are properties of phrases, not individual words, and that their semantic interpretation is therefore defined over phrasal expressions. This brings inflections on lexical heads into line with all other exponents of functional features.

1. The lexeme concept

All theories of morphosyntax appeal to a notion of 'lexical entry', a representation of formal and conceptual properties of a word, distinguishing it from other words and capturing those unpredictable properties which govern its morphological and syntactic behaviour. For morphologists who adopt a 'realization' approach to inflection (and possibly derivation), the lexical entry is the

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'lexeme'. A standard characterization of 'lexeme' following the tradition inaugurated by Matthews (1972) and developed by Aronoff (1994), Beard (1995) and others would run something like (1):

- (1) Lexeme: Set of words forms united by constant meaning

This characterization captures two important claims: (i) a lexeme has a single meaning, (ii) the lexeme generalizes over inflected forms. This notion of lexeme hinges on the distinction between inflected forms of words (grammatical words), as opposed to derived lexemes. That is, the lexeme concept presupposes the distinction between inflection and derivation. A derivational process will create a new lexeme, which minimally ought to mean that it adds some element of meaning to the base lexeme.¹ If the basic difference between lexemes lies in conceptual meaning then it will be important to distinguish between the meanings conveyed by inflections and those conveyed by derivation. For instance, we do not want to say that the form *dogs* realizes a different lexeme from the form *dog* on the grounds that *dog* means [DOG] while *dogs* means [PLURAL[DOG]]. Yet it is a common assumption that inflections add meaning to words. Anderson (1992, 79) explicitly states that *dog* and *dogs* have different meanings, and Booij (1994; 1996) draws a distinction between inherent inflections, those inflections which bear meanings in themselves such as Number or Tense, and contextual inflections, those inflections which are purely formal and are determined purely by the syntactic context (agreement, government). It is clear from Booij's characterization that inherent inflections are properties of the word.

If inflected words may bear the meaning of those inflections, then just looking at the semantics of a word will not tell us whether a morphological process that affects it is inflectional or derivational. Clearly we need a way of distinguishing the properties of inflections from those of derivations. A checklist of such properties is provided by Plank (1994). However, he argues that the distinction is, in fact, gradient (cf. Bybee 1985; Haspelmath 1996). For Anderson (1992) inflection and derivation are handled in different components

¹ In fact, the discussion of derivation here is a massive oversimplification, since there is derivation which is entirely divorced from semantic interpretation as well as morphology which appears to be derivational but which adds no new meaning, i.e., transpositions such as participles, action nominalizations and gerunds ('masdars') (see Haspelmath 1996) and also relational adjectives (*morphology* ~ *morphological*), and perhaps property nominalizations of adjectives (*sweet* ~ *sweetness*). See Spencer (1999) for proposals for treating transpositions.

of grammar (syntax and the lexicon respectively), and we will see that this is an important insight. However, this solution still leaves open the problem of how we know (or rather, how the language learner knows) that [PLURAL[DOG]] represents a word form of a single lexeme DOG and not a distinct lexeme (created by the syntax).

It is important to understand that inflectional morphology is just one way in which a language realizes its f(unctional) features. I shall assume that every language has a vocabulary of f-features, which have two main roles: (i) to regulate syntactic structure (e.g. agreement, government) and (ii) to express certain 'functional' meanings, such as Tense, Definiteness, Plural, etc. In the context of inflection these two roles correspond respectively to contextual inflection and inherent inflection in the sense of Booij (1994; 1996). F-features may be expressed in a great variety of ways apart from inflection of the head word, including clitics, auxiliary words, word order and intonation, or by a combination of these. Intonation extends over phrases and presumably can only be interpreted on the basis of a parsing of the intonation phrase. Word order by definition is defined over sequences of words. In many syntactic frameworks auxiliary elements such as auxiliary verbs, determiners, degree modifiers and so on are regarded as heads of functional phrases which take lexically headed phrases as complements. In those theories, the semantic interpretation of auxiliary words must therefore by definition include the lexical phrase in its scope. Clitics and edge inflections are generally defined with respect to a position at the periphery of a phrase. For instance, the possessive *-s* in English appears at the right edge of a determiner phrase, even though it relates semantically to the lexical head of the noun phrase (e.g., [*the man we were talking to*]'s] *name*). Such cases of phrasal affixation are generally analysed as peripheral marking serving as the exponent of a phrase level feature (e.g., Halpern 1995). This means that if head-inflections such as Plural or Past are interpreted at the word level rather than the phrasal level they are unique amongst f-features.

2. Inflection and the architecture of the lexicon

In any theory it is necessary to establish a correspondence between syntactic f-features and (i) the realization of those features; (ii) the semantic interpretation of those features. In many cases there is a reasonably clear correlation between morphology and syntactic f-features. Thus, if we regard English modals as exponents of mood f-features we can say that there is a feature,

Analytic constructions often provide a subtle exemplification of this. Consider the English perfect aspect/tense: *Tom has eaten the apple*. Here the syntactic feature [ASPECT:Perfect] is expressed by a combination of auxiliary *have* and *-en* participle. Both of these have other functions (*have* is also used as a modal auxiliary of obligation as well as a meaningless light verb as in *have a bath*). Thus, the morphosyntax has to spell out the aspectual feature by reference to an auxiliary and a participle form. In languages in which the perfect has come to be used to mean simple past the mismatch between syntactic and morphological features is more apparent. In the Colloquial French passé composé *Tom a mangé la pomme* ‘Tom has eaten the apple’, the feature [TENSE:Past] is conjointly realized by the morphological functions PRES(*avoir*) + past participle. In other words, we need to adopt Separationism (cf. Beard 1995) at the interface between morphology and syntax. (See Ackerman–Webelhuth 1998 for detailed discussion of the implications of such cases).

We therefore need at least three sets of representations to account for the realization of f-features. Here I shall adopt a modification of the architecture for lexical entries proposed by Jackendoff (1997), in which an inflected word form reflects a coindexation between three sets of representations (see Spencer 1997 for more detailed discussion of Jackendoff's proposals). The first is the output of the morphological component (here conceived of as a set of Paradigm Functions, after Stump 1993). This component operates autonomously (cf. Aronoff's (1994) notion of morphology-by-itself, an instantiation of Separationism). The second is a syntactic representation and the third a semantic representation. In (2) we see the entry for *dog*:

- $$(2) \quad {}_1\text{dog} \quad {}_1\text{N}_1 \quad [\text{Thing}^{\text{DOG}}]_1$$

In (2) we see that the syntactic category label is given an identical right and a left subscript cross-referencing a semantic and a morphological representation respectively. However, this only accounts for an underived, uninflected lexical entry. Now consider the inflected forms, specifically the plural. In a realization theory the plural would be specified by a function or set of functions applying to the lexeme (cf. Aronoff 1994) or to the root of the lexeme (Stump's (1993) Paradigm Functions). Call the function which delivers plural forms PLUR. Suppose that this function is coindexed with a syntactic feature [NUM:Pl]. Using Jackendoff's notation we could represent this as in (3):

$$(3) \quad \begin{array}{ccc} {}_{10}\text{PLUR}({}_1\text{dog}) & {}_{10}[\text{NUM:Pl}]_{10}({}_1[\text{N}]_1) & [{}_{\text{Thing}}\text{PLUR}[{}_{\text{Thing}}\text{DOG}]_1]_{10} \\ | & & \\ \text{dog}_z & & \end{array}$$

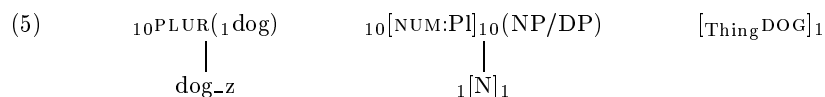
I have taken the liberty of interpreting the functional feature here as a function applied to the lexical category feature in the syntax, to mirror the functional application in the morphological and semantic representations.

But as we know, (3) is odd because it seems to be saying that inflection creates a new lexeme. Given the logic of the lexeme concept we should really be assuming the set of representations in (4):

$$(4) \quad \begin{array}{ccc} {}_{10}\text{PLUR}({}_1\text{dog}) & {}_{10}[\text{NUM:Pl}]_{10}({}_1[\text{N}]_1) & [{}_{\text{Thing}}\text{DOG}]_1 \\ | & & \\ \text{dog}_z & & \end{array}$$

In other words, *dogs* is a form of the DOG lexeme which bears an exponent which is (somehow, somewhere) interpreted as 'plural'. Note that we need representations of this sort anyway for Booij's contextual inflections. Given (4), *dogs* differs minimally from *the dog*, which is also a morphosyntactic expression that contains an exponent of the DOG lexeme (note, not the THE DOG lexeme!) in addition to an exponent of the feature 'definite'. The difference between the plural and the definite form is simply that the plural is expressed by means of a Paradigm Function applied to the root of the lexeme, while the definite form is expressed by means of an auxiliary element (definite determiner). In languages with definiteness affixes rather than definite articles the parallel between a definite dog and a plural dog would be complete. Semantically speaking, plurality cannot be a property of a bare common noun but only of a referential element. The syntactic unit corresponding to a referential element is a noun (or determiner) phrase. The obvious way to capture this is to say that the syntactic plural feature is a property of the noun phrase, or

perhaps even determiner phrase, where it receives its semantic interpretation (*pace* Anderson 1992). This view of the word form is shown in (5):



The meaning of the plural marker (or indeed, whether it gets interpreted semantically in the first place) depends on the wider syntactic context. The correspondence rules ensure that the PLUR function in the morphology licenses the appearance of the [NUM:Pl] feature on the noun/determiner phrase in the syntax. If nothing else is said, the coindexation will ensure that the word form which realizes the [NUM:Pl] feature is the lexical head (other machinery is necessary to account for edge inflections, as argued by Halpern 1995).

There are several reasons for marking and interpreting inflections at the phrasal level only.

First, it brings head-inflections into line with all other exponents of f-features (including edge inflections).

Second, there are occasions when it is quite unclear what the additional meaning component of *dogs* actually is, if it exists at all, as in the expression *two dogs*. The simplest representation for this will be something along the lines of $[2x[\text{DOG}(x)]]$, without any semantic interpretation given to the syntactic plural feature, which is therefore interpreted as a kind of agreement. Not all languages insist on this type of agreement, of course (e.g., Hungarian), and in English agreement is not found with nouns denoting game animals (*We bagged a brace of pheasant/*pheasants*). In point of fact, it might be embarrassing if the plural marker were given semantic interpretation in *two dogs*, because then the expression ought to mean ‘two groups of dogs’. Now, there are no doubt ways of defining the semantics of plurality in such a way as to avoid this (see Ortmann 1998, for instance), but if the syntactic feature is phrasal and if the syntax-semantic correspondence principles recognize it as an agreement, the simplest interpretation of plurality can be maintained.

There are other constructions in which the plural fails to receive semantic interpretation. Thus, in (6a, b) we have two different ways of saying essentially the same thing:

- (6) (a) these types of dogs
 (b) these types of dog
 (c) this type of dog
 (d) *this type of dogs

In (6a, c) it appears as if there is some sort of number agreement between *type* and *dog*, which is not, however, obligatory ((6b) with a plural head noun). Likewise, in predicative constructions such as (7) we find number agreement:

- (7) (a) None of her daughters were doctors
 (b) None of her daughters was a doctor
 (c) *None of her daughters were a doctor
 (d) *None of her daughters was doctors

In (7a, b) we see that *none* can be treated as singular or plural, but once that decision is made it must be carried through for the whole predicate (cf. (7c, d)). However, since (7a, b) are synonymous it can hardly be the case that the plural morphology contributes meaning (to the lexeme or to anything else).

The past tense inflection of English verbs is if anything even more problematical, for well-known reasons. Thus, given sequence of tenses, we don't want to have to say that *arrived* in (8) means [PAST[ARRIVE(*x*)]]:

- (8) I thought you arrived tomorrow.

Here, the past tense marking is conditioned by agreement, and thus constitutes a semantically uninterpreted morphosyntactic feature. The last thing we want *arrived* to mean in (8) is [PAST[ARRIVE(*x*)]].

Constructions such as the English perfect tense pose interesting problems when we come to ask about semantic interpretation. In an expression like *Tom has eaten the apple* the meaning of 'perfect' is expressed neither by *has* nor by *eaten*. This is because the participle is generally ambiguous between the perfect and the passive reading (at least for transitive verbs) and *have* is ambiguous between a main verb, a modal auxiliary (*Tom has to leave*), and, arguably, a meaningless 'light' verb (*Tom had a bath*). What conveys the meaning 'perfect' is the combination of auxiliary and participle. This is different from a case such as *Tom must leave* in which we can identify the modal force of the sentence simply with the auxiliary. We can think of the modal auxiliary as an example of a compositional analytical construction, and the perfect tense as a non-compositional analytical construction. An interesting question (examined at

some length in Ackerman–Webelhuth 1998) is the relationship between such analytical constructions on the one hand and synthetic constructions like the simple past (*Tom ate an apple*) on the other. In one sense, these are in a paradigmatic relation to each other, and the perfect or progressive forms are part of the inflectional paradigm of the verbal lexeme. This is particularly obvious in cases of periphrastic inflection of the kind discussed for Latin by Börjars–Vincent–Chapman (1997).

The reason this is of interest is that it is very difficult to see how to state the paradigmatic relationship between synthetic constructions and analytic constructions given the standard conception of inflection. This is because classical inflection is only a part of the story. A full grammar of English has to set correspondence rules between syntactic features such as [ASPECT:Perfect, TENSE:Present] and ‘*have* + past participle’ constructions. The correspondence rules which tell us that the morphological function PAST(*leave*) (i.e., *left*) corresponds to the syntactic feature [TENSE:Past] applied to LEAVE will also tell us that *has left* is the construction required to realize the syntactic feature specification [ASPECT:Perfect, TENSE:Present].

There remains one question: how does the grammar differentiate between uninterpreted inflections and interpreted inflections? Note that this is a problem for any theory of grammar, though not one which has received much discussion. (Booij (1994; 1996) does not explicitly address the problem, for instance, in his discussion of contextual and inherent inflection.) The solution depends on the overall architecture of the grammar. One suggestion we might make is the following. Suppose we follow Lexical Functional Grammar (Bresnan, in press) in assuming a level of c-structure defining the phrasal constituency of an expression and a level of f-structure at which functional properties such as grammatical relations, tense and so on are expressed. In standard LFG, this level also contains feature specifications governing agreement and government (contextual inflection). However, suppose we say that all inherent f-features are represented twice, once in f-structure and also as node labels at c-structure. Suppose, too, that contextual features are represented only at c-structure, being distributed there by specific rules of formal agreement and government. Finally, suppose that all semantic interpretation is defined over f-structure (as in standard LFG). In this fashion we can make a simple architectural distinction between the two types of feature. This is not the standard practice in LFG, though it is not incompatible with the basic precepts of the theory as far as I can tell.

3. Conclusions

The model argued for here shares much with Anderson's (1992) conception of 'split morphology'. However, Anderson argues for an architecture in which inflections are actually specified in the syntax. In my model, derived from the perspective on the lexicon offered in Jackendoff 1997, it is possible to accommodate all morphology, derivational or inflectional, in an autonomous morphological module, while still enjoying the benefits of split morphology. I adopt Jackendoff's suggestion that the components of a word are linked to each other by means of a set of lexical indices. In simple cases there is a one-one correspondence between features at different levels, though in more complex cases we might find a more complex relationship (as in the case of the English perfect or Colloquial French *passé composé*). The split between inflection and derivation is apparent from the correspondence patterns of features at various levels of representation. A piece of morphology will be interpreted as inflectional if and only if it corresponds to a syntactic feature. Syntactic features, if interpreted at all, are interpreted at the phrasal level. This helps explain why inherent inflections such as the English plural sometimes behave as though they were contextual inflections, in not receiving a semantic interpretation. In this way inflections are seen to be no different from other functional features (though we are not obliged to say that inflections are syntactic heads projecting their own phrases, as in most Principles and Parameters models of syntax). On the other hand a piece of morphology is derivational if it is given direct semantic interpretation at the lexemic level.

As pointed out by opponents of split morphology, there is no formal difference between morphological operations subserving inflection and those subserving derivation (indeed, one and the same operation can subserve both at the same time for the same lexeme, as we find with, say verbal and adjectival passives in English). However, this is expected in a theory which countenances an autonomous morphological module and the principle of Separationism, as here.

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